Experiment 1.1

Generation and Classification of Continuous-Time and Discrete-Time Signals

Objective

(i) To generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal, and exponential signals.

(ii) To generate and visualize basic discrete-time signals and analyze their amplitude and timeindex characteristics.

Expected Learning Outcomes

• Generate impulse, step, ramp, exponential and sinusoidal continuous-time signals.

• Obtain correctly scaled time-domain plots.

• Determine measurable parameters with preferably ≤5% deviation where applicable.

• Explain the generated waveform using its mathematical expression.

To verify the generation of a sinusoidal signal with a specific frequency, amplitude, and phase

shift using MATLAB. The Aim is to ensure that the signal generator can accurately produce a

sinusoidal waveform with the desired parameters.

Software / Laboratory Environment

MATLAB

Formulae / Analysis

$x(t) = A \sin(2\pi f_0 t + \varphi)$
$T_0 = 1/f_0$ for a sinusoidal signal.
$x(t) = e^{(-at)}\, u(t)$ for a decaying exponential.
Continuous-time signals are defined for every value of t.
Percentage deviation = |Practical − Theoretical| / |Theoretical| × 100

Experimental Procedure

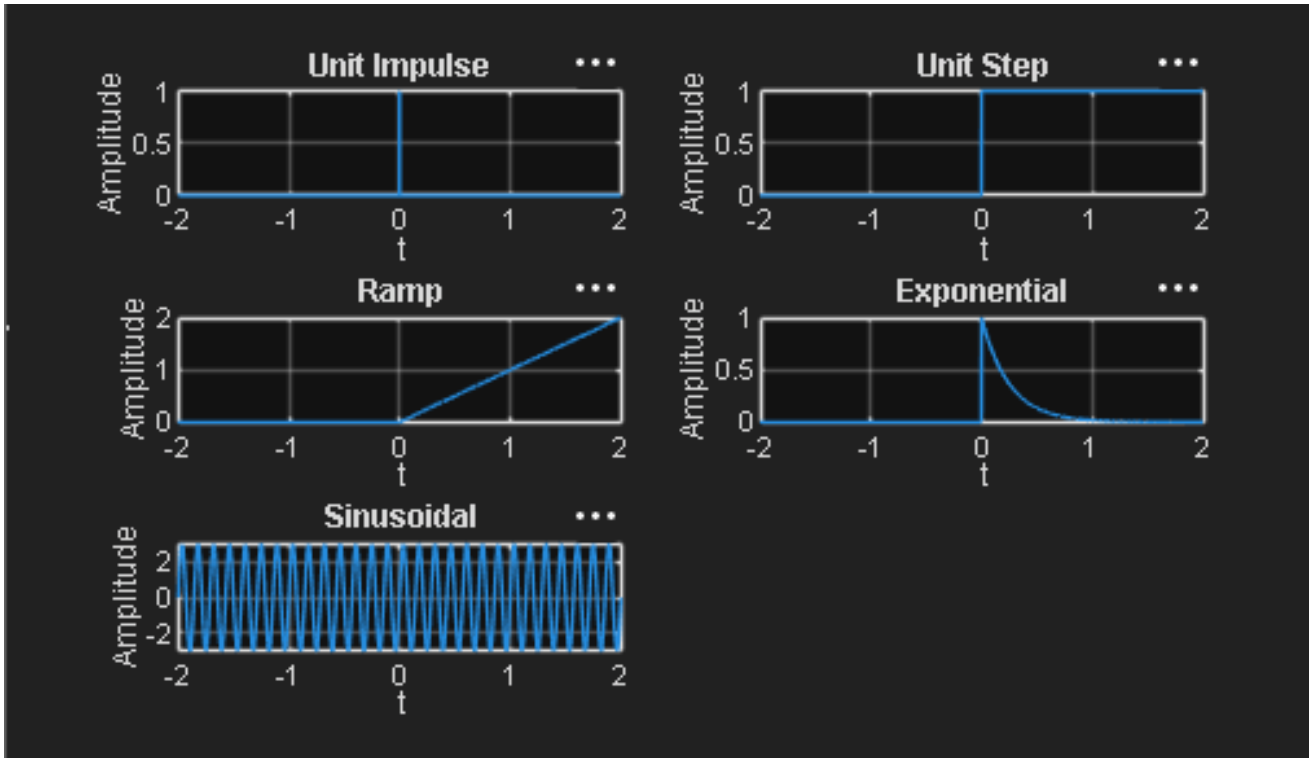
1. Define a continuous time vector with a sufficiently small simulation interval.
2. Generate unit impulse, unit step, ramp, exponential and sinusoidal signals.
3. Plot each waveform against time using suitable amplitude and time scales.
4. Measure amplitude, frequency and period wherever applicable.
5. Compare the simulated waveform with its theoretical expression.
6. Record observations and justify the signal characteristics.

Key Conditions & Threshold Values for Analysis

- Use a sufficiently small simulation interval to represent the waveform accurately.
- The time axis must clearly display the relevant waveform features.
- For a sinusoidal signal, $T_0 = 1/f_0$.
- Numerical deviation for measurable parameters should preferably be $\leq 5\%$.

MATLAB Coding

```
% Experiment 1: Generation of Continuous-Time Signals
 clc; clear; close all;
 t = -1:0.001:1;
 A = 2; f0 = 5; a = 1;
 imp = double(abs(t)<0.0005);
 step = double(t>=0);
 ramp = t.*step;
 expSig = exp(-a*t).*step;
 sinSig = A*sin(2*pi*f0*t);
 figure;
 subplot(3,2,1); plot(t,imp); grid on; title('Unit Impulse'); xlabel('t');
ylabel('Amplitude');
 subplot(3,2,2); plot(t,step); grid on; title('Unit Step'); xlabel('t');
ylabel('Amplitude');
 subplot(3,2,3); plot(t,ramp); grid on; title('Ramp'); xlabel('t'); ylabel('Amplitude');
 subplot(3,2,4); plot(t,expSig); grid on; title('Exponential'); xlabel('t');
ylabel('Amplitude');
subplot(3,2,5); plot(t,sinSig); grid on; title('Sinusoidal'); xlabel('t');
ylabel('Amplitude');
```



RESULT

The continuous-time and discrete-time signals were generated and analyzed successfully. The signals were classified as periodic or aperiodic, and the even and odd components were obtained and used to reconstruct the original signal successfully.

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
1. Generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal and exponential.	**CT Signal Generation & MATLAB Implementation**	Generates all five CT signals accurately with a suitable time vector and step size. Uses correct MATLAB commands . Plots are correctly scaled, with titles, axis labels and grid.	Generates the required CT signals and plots correctly, with minor errors in code, time range, parameters or labelling.	Generates basic CT signals but has noticeable errors in code, step or impulse representation, or plots, and needs assistance.	Unable to generate the required CT signals or produce meaningful plots.	**25**	**2.5**	
2. Generate and visualize basic discrete-time signals such as unit sample, unit step, ramp, sinusoidal and exponential sequences.	**DT Signal Generation & MATLAB Implementation**	Generates all five DT sequences accurately with a proper index vector n. Uses stem correctly. Plots are correctly scaled, with titles, axis labels and grid.	Generates the required DT sequences and plots correctly, with minor errors in index range, stem usage, parameters or labelling.	Generates basic DT sequences but has noticeable errors in code, indexing or stem plots, and needs assistance.	Unable to generate the required DT sequences or produce meaningful stem plots.	**25**	**2.5**	

3. Classify CT and DT signals as periodic or aperiodic and identify their waveform characteristics.	**Signal Classification & Periodicity Analysis**	Correctly classifies all signals using DT. Determines the period where applicable. Identifies frequency, phase, and growth or decay. Explains the reasoning clearly with mathematical conditions and waveform evidence.	Classifies most signals correctly and finds most periods, with minor errors or limited justification.	Gives basic classifications. Period calculation or justification is incomplete and needs guidance.	Unable to classify the signals correctly or justify the classification.	**25**	**2.5**	
4. Analyze the amplitude and time-index characteristics of discrete-time signals.	**Amplitude & Time-Index Analysis**	Correctly determines maximum and minimum amplitude, the index n at which each occurs, and signal energy. Correctly shows the effect of time shifting and folding on the index. Interprets the results clearly.	Analyzes most amplitude and index characteristics correctly, with minor errors in values or interpretation.	Gives basic amplitude and index analysis with noticeable errors, and needs guidance.	Unable to analyze the amplitude and time-index characteristics or interpret the results.	**25**	**2.5**	
					Total	**100**	**10**	